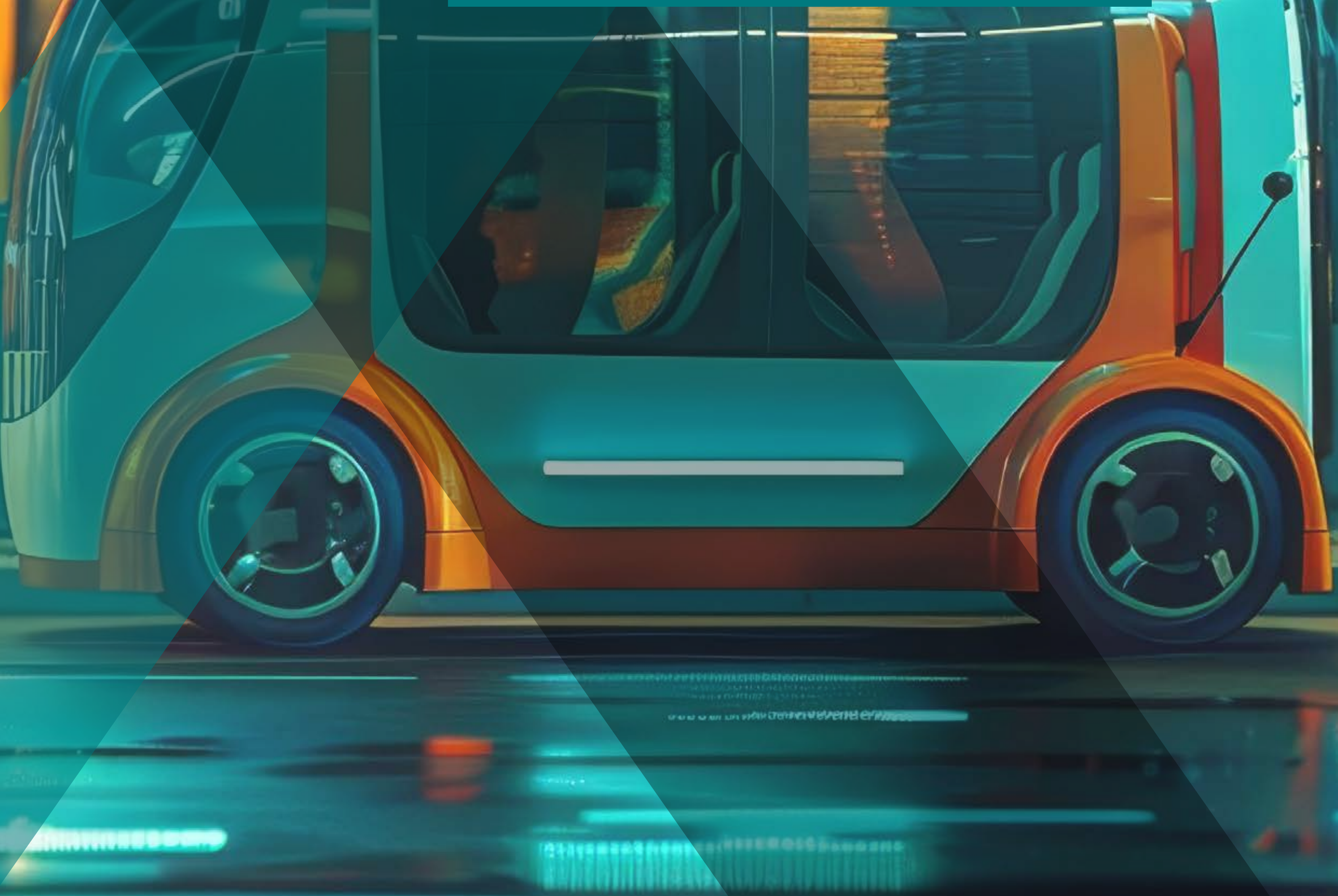
The Future of Driving is Autonomous

Strategy of the Federal Government for Autonomous Driving in Road Traffic



The Federal Government

Foreword

Germany is becoming increasingly digital – and so is its transport and mobility sector. Autonomous and connected driving will play a key role in this sector's future. We believe in this technology. With the present Strategy, we want to advance it and leverage the immense opportunities it offers. And that's not all. We want to make Germany one of the world's leading hubs for innovation and production in the autonomous driving sector.

In the future, autonomous driving is to be a common sight in normal operations throughout Germany. We are working towards this goal, focusing above all on local public transport and freight transport. These uses in particular can deliver vital impetus for innovative, autonomous mobility concepts. They also build trust and acceptance: The more autonomous vehicles on our roads giving people a positive impression of this innovative technology in everyday life, the clearer its benefits will become and the faster the enthusiasm for it will grow.

Take local transport for example: Autonomous vehicles can increase participation and improve the quality of life – especially for people in rural areas, where seamless local transport systems often struggle to remain economically viable. Anyone who cannot or does not want to drive their own vehicle will be able to conveniently have an autonomous shuttle bus pick them up and take them to their destination.

We want to make this vision a reality. The Federal Government is paving the way to that goal with this Strategy. It identifies measures and recommends actions to accelerate the growth of the market for autonomous driving. While the Strategy focuses on road traffic, it will also touch on the rail, waterways and aviation modes, as autonomous systems offer major opportunities in these sectors, too.

We are creating a framework conducive to innovation, one which will make mobility more autonomous and connected in the future. It is up to industry and the private sector to breathe life into this framework. Together, we will succeed in making autonomous vehicles a common sight on our roads.

Dr Volker Wissing,
Member of the German Bundestag
Federal Minister for Digital and Transport

Table of contents

1 Guiding principles of the Strategy

Technologies related to autonomous and connected driving will revolutionize mobility. They can help increase traffic safety, boost local public transport, improve connections to rural areas, remote regions and districts further from the city centre, and enhance the quality of life – including that of persons with mobility constraints. They could also help alleviate future shortages of professional drivers in passenger and freight transport, and thus, in the long term, contribute to maintaining a comprehensive range of mobility services.

As early as 2015, the Strategy for Automated and Connected Driving[1] addressed the relevant opportunities and challenges. A report on the implementation of the Strategy was published in 2017[2]. It found that the Strategy had delivered essential impetus and laid key foundations. In 2021, Germany introduced the world's first rules for the use of autonomous motor vehicles in normal operations on public roads in its regulatory framework for autonomous driving. Many research projects have successfully tested autonomous driving systems in various use cases.

By refocusing the Strategy, the Federal Government aims to enhance Germany's status as an innovator in the autonomous driving segment and thus breathe new life into the existing regulatory framework for autonomous driving[3]. At the same time, the Federal Government is committed to leveraging the potential to influence the intelligent mobility of tomorrow and addressing the challenges and risks appropriately.

The Strategy identifies measures and proposes actions to further improve the framework conditions to ramp-up the market for autonomous driving. There is potential for promising applications of autonomous driving in both local public transport and freight transport. We consider local public transport an important area of application and a driving force for innovative, autonomous mobility services. We aim to be world leading in autonomous driving within the freight transport sector.

With its holistic Strategy, the Federal Government is also continuing to promote the capacity of the German automotive industry and its component suppliers to innovate and keep pace with the global technology competition. This also strengthens Germany's status as an industrial and business-based economy.

While the Strategy focuses primarily on road traffic applications, it also considers use cases in other modes of transport as examples, to highlight the technology's intermodal importance along the mobility chain and the

1 BMDV (2015): https://bmdv.bund.de/SharedDocs/EN/publications/strategy-for-automated-and-connected-driving.pdf?__blob=publicationFile

2 BMDV (2017): https://bmdv.bund.de/SharedDocs/DE/Publikationen/DG/bericht-avf.pdf?__blob=publicationFile (in German only)

3 The applicable regulatory framework refers to SAE level 4 (see page 8 for a definition), which is also covered by this Strategy.

potential for innovation. The Strategy's contents, especially the action areas it derives, are also transferrable. It emphasizes their significance for other modes of transport against the background of a future-proof and resilient transport system.

The Federal Government is basing its approach on the following guiding principles:

- We are promoting the rollout of autonomous driving systems in normal operations[4] nationwide, focusing in particular on local public transport and freight transport.

- We are pursuing a sustainable and efficient deployment of autonomous and connected transport systems and putting in place the general frameworks to leverage potential.

- We are increasingly working to put in place organizational and operational frameworks in order to facilitate the application of autonomous and connected driving also across federal state borders on federal territory, for example on federal motorways and federal highways.

- We will enhance the innovative capacity of the German automotive industry by evolving the frameworks for upscaling of market-ready technologies.

- We will boost the uptake of autonomous and connected mobility services so that as many people as possible use these innovative technologies in public transport and are enthusiastic about them.

- We are enhancing our dialogue with citizens in order to inform the planning and rollout of autonomous transport systems with their perspectives. This can help to boost societal acceptance and the population's adoption of the technology.

- We are promoting transport users' trust in autonomous and connected driving by reviewing and updating the general frameworks for the systems' road safety and cyber security on an ongoing basis.

- We are relying on commercial investments and will, where possible, provide targeted support for the development, demonstration and market launch of autonomous mobility services wherever this is in the public interest.

- We are championing coherent and complementary promotion of research and development by the Federation and the federal states for greater competitiveness and faster scaling of applications. Public-sector research and development activities and Federal research projects are being evaluated as a basis for further activities on autonomous driving, after which the results are published.

- We support continuous exchanges of information and experience between industry, the scientific community, research establishments, authorities and international partners in Germany and abroad to benefit from best practices for innovative and efficient processes.

4 Normal operations are defined as a state in which autonomous driving on public roads is integrated as a permanent, nationwide standard service beyond the scope of project and test operations.

- We want to avoid potential rebound effects caused, for example, by the increased convenience of autonomous driving leading to more private journeys. New mobility services, such as driverless transport services and digital platforms for such services, must permit meaningful integration in mobility systems. We are funding studies to this effect and putting in place the necessary frameworks.

- We want to ensure that the development and deployment of the new technologies will not hinder the opportunities for environmental protection and climate action – such as reducing emissions – in the future.

- We will be focusing on improving road safety in all aspects.

The measures revealed in this Strategy take into account actions recommended by the Round Table on Autonomous Driving and the Expert Group on the Transformation of the Automotive Industry. As independent bodies, they advise the Federal Ministry for Digital and Transport and the Federal Ministry for Economic Affairs and Climate Action.

Where specific measures give rise to federal budget expenditures, they are subject to the availability of funds and must be financed from savings in other areas within the current budgetary and fiscal planning of the appropriate government departments.

2 Status quo

2.1 Levels of automation

In the road mode, six levels of automation are distinguished. They were initially drawn up by the Federal Highway Research Institute (BASt) and have been adopted by SAE International (formerly the Society of Automotive Engineers).

- **Level 0:** No features with a continuous effect on driving.
- **Level 1** and **2: Driver assistance systems** that help drivers drive the vehicle safely.
- **Level 3:** Vehicles with **automated driving systems** that take over the complete driving task autonomously. However, drivers must always be ready to take control of the vehicle again immediately if required by the system.
- **Level 4:** In specific operational areas, vehicles can perform the complete driving task without drivers being required to intervene in the control of the vehicle (passenger role). **Autonomous driving** is said to begin at Level 4.
- **Level 5:** Vehicles with **Level 5 systems** can drive anywhere a driver with normal training can typically drive appropriately. The Strategy does not cover Level 5 systems.

To facilitate user communication, the terms assisted, automated and autonomous driving were introduced and will be used below (see the figure below).

Driving mode	Assisted	Automated	Autonomous
SAE level	1 and 2	3	4 and 5
User-vehicle interaction	Driver		System
	– The driver must monitor the system and environment continuously. – The driver must take corrective action. – The driver is responsible. – The system helps the driver complete the driving task.	– Complete transfer of control to the system on activation. – The driver can perform other tasks. – If requested by the system or in the event of obvious anomalies, the driver must take control.	– After activation and, where applicable, within a defined operational area, the system performs the driving task. – The operator can deactivate the system. – In certain SAE Level 4 and 5 systems, passengers may have the option of ending autonomous driving. – If provided for and possible, a driver can take control.
Examples	Assistance systems such as: – Adaptive cruise control – Lane keeping assist	Automated control systems such as: – Automated Lane Keeping System on motorways (ALKS)	Mobility services such as: – Driverless bus/shuttle services – Robotaxis – Autonomous hub-to-hub services

Driver performs the driving task

System performs the driving task

Overview of assisted, automated and autonomous driving (own illustration based on Federal Highway Research Institute (BASt))

2.2 Achievements

Milestones of autonomous driving in Germany

In its 2015 Strategy for Automated and Connected Driving, the Federal Government formulated key guidelines and principles to harness the potential of automated and connected driving. The Strategy underpins the Federal Republic's role as an international catalyst.

At a global level, the Federal Republic of Germany is considered a pioneer in the creation of the legal prerequisites. The Act on Automated Driving (2017), the Act on Autonomous Driving (2021) and the AFGBV (2022) make Germany the first country in the world to have fully put in place a statutory basis for autonomous driving (Level 4). As early as 2017, the Ethics Commissions formulated guidance for automated and connected driving. Guidelines on assessment, including requirements and evaluation criteria for the authorization of defined operational areas, have also been available since 2024.

With regard to infrastructure requirements for autonomous driving, the Federal Autobahn GmbH is currently equipping all mobile road works warning signs at daytime roadworks with the new digital communication system (C-ITS). This enables them to send timely and reliable warnings of daytime roadworks also to connected vehicles.

The Federal Ministry for Digital and Transport is supporting **ongoing projects in real world conditions** based on funding guidelines, focusing on integration into normal operations.

Federal Government research and innovation promotion plays a key part in trialling the technology in test and real world conditions. The Federal Ministry for Digital and Transport, the Federal Ministry for Economic Affairs and Climate Action and the Federal Ministry of Education and Research have distributed €642 million in funding since 2016. Innovative technologies have also been trialled at no fewer than 26 test beds. Since 2019, the Research for Autonomous Driving Action Plan has provided a common framework for research funding.

German industry remains a world leader in vehicle automation. In late 2021, the world's first type-approval was granted for an Automated Lane Keeping System at 60 km/h. In 2024, the first type-approval up to a speed of 95 km/h was issued under specific conditions. First Automated Parking Systems (APS) have been operating autonomously since 2022.

Milestones in road traffic

With regard to the rollout of autonomous driving, Germany has already gained valuable insights and set key measures in motion:

- The national regulatory framework[5] makes Germany a global pioneer when it comes to regulating automated (Level 3) and autonomous driving (Level 4). As a result, autonomous motor vehicles can already be operated in defined operational areas on public roads in normal operations.
- Vehicle and system testing for different use cases is being developed, and deployment in real-world driving conditions on the roads is being supported through targeted promotion of research and development for the necessary underlying technologies and promotion of test operation.
- The Federal Autobahn GmbH is responsible for the fourth longest motorway network in the world. In 2023, a first Cooperative Intelligent Transport System (C-ITS) was introduced for direct communication between infrastructure and vehicles: the road works warning system.

5 The Eighth Act amending the Road Traffic Act (of 2017), the Act amending the Road Traffic Act and the Compulsory Insurance Act (Act on Autonomous Driving of 2021), the supplementary Ordinance on the Approval and Operation of Motor Vehicles with Autonomous Driving Functions in Defined Operational Areas (AFGBV of 2022) and the Carriage of Passengers Amendment Act (PBefG) of 2021 form a legal basis for the integration of on-demand pooling in the local public transport system.

- C-ITS services are being trialled for traffic flow management in several cities. They help enhance road safety.

- In recent years, test beds were established for automated and connected driving. They were used to implement research projects and to evolve and trial technologies and existing systems.

- German automotive manufacturers have made significant progress and are at the global forefront in the development levels up to and including autonomous driving, especially for automated driving systems. For example, under certain circumstances, automated driving systems are permitted to take over the complete driving task and make driving safer and more convenient at speeds of up to 130 km/h on motorways.

- Key national stakeholders in the automated driving sector have formed networks to advance the rollout of automated driving technologies. In recent years, a range of communication and coordination channels have established themselves, such as the Round Table on Autonomous Driving and the Expert Group on Transformation in the Automotive Industry. They are supplemented by regular events, including the Research and Technology for Autonomous Driving Conference.

A look at other modes of transport

- The regulatory framework for fully automated rail transport is also highly advanced, but primarily with regard to closed metro systems. Driverless underground trains have been in operation in Nuremberg since 2008, and can run twice as often as with manual control. The employees in the control centre only intervene in emergencies.

- As part of the Federal Government Programme on the Future of Rail Freight, projects for the testing and market launch of innovative future technologies in rail freight transport in digitalization, automation and vehicle technology are being funded. This also includes, for example, projects to test automated freight trains.

- The following measures have been taken to promote autonomous navigation on inland waterways:

  - The Digital Test Beds on Waterways financial assistance programme enables the industry to test and evolve systems up to and including fully automated navigation. For example, applications in autonomous ferries and classic inland navigation are being investigated, as well as alternative transport concepts in urban logistics and shuttle systems.

  - The regulatory framework has been expanded to include experimental clauses that allow innovative technologies – such as remote-controlled inland navigation vessels – to be tested on the water.

- In maritime shipping, a regulatory framework for EU-wide data exchange was created in 2019 for the use of Union-wide harmonized digital maritime transport systems and maritime transport services as well as connected IT systems.

- Since 2016, the Federal Ministry for Digital and Transport (BMDV) has been funding R&D projects related to digital data-based applications for mobility of the future with an innovation initiative called mFUND. Many research projects for unmanned aerial vehicles and electric Vertical Take-Off and Landing Aircraft (eVTOL) benefit from the funding, particularly with regard to the safe integration of new aircraft into existing airspace structures.

2.3 *Road mode*

2.3.1 Private motorized transport

More and more people rely on and use vehicles with assistance systems. The availability of automated driving features is also on the rise. This includes applications that can take over the driving task in certain driving situations, such as on congested motorways at speeds of up to 60 km/h and, increasing progressively in stages, at up to 130 km/h. There are also already autonomous features for parking (Automated Parking Systems, APS). A gradual rollout has the advantage of allowing the technical maturity of the systems to be tested and gradually introducing users to autonomous driving and boosting their trust in the technology through personal experience in specific driving situations.

There are already measures to support autonomous driving at a variety of levels. For example, individual variable traffic signs on federal motorways are being retrofitted with cooperative intelligent transport systems to trial the direct transmission of warnings and dynamic speed limits to appropriately equipped vehicles. This makes an exchange of information between traffic management centres and vehicles possible.

2.3.2 Local public transport

Given its high space and energy efficiency, as well as its social welfare and participation function, local public transport is a key action area for sustainable mobility. It has the potential to become a driver of autonomous systems. Given the shortages of driving staff, these systems can play a key role in the long term.

On-demand shuttles and potential autonomous regular scheduled buses make new public transport services possible. A transport system that supplements busy traffic with responsive on-demand services (especially as feeders) contributes to more efficient mobility and can deliver better services, in particular for rural areas, remote regions and districts further from the city centre. On-demand services offer alternatives to private passenger cars, can make a long-term switch to public mobility services more attractive and help increase the use and capacity utilization of local public transport.

Technical solutions are currently being developed and the use of autonomous shuttles and buses to extend the timetables and geographical reach of local public transport services is being successfully trialled in a series of funding projects at federal and federal state levels.

Operation in particular of autonomous on-demand shuttles is currently in a phase between the end of trialling and the start of upscaling. A trend of German providers withdrawing from the market and withholding investments is apparent. At present, there is an absence of both autonomous shuttle vehicles ready for mass production and scalable operator and business models. This is currently making it difficult to profitably deploy autonomous on-demand shuttles, with the exception of geographically limited individual solutions.

KIRA

Use case: KIRA project – Development of an autonomous on-demand fleet

In the KIRA project, Rhein-Main-Verkehrsverbund GmbH and Deutsche Bahn AG, together with other partners, are working on an on-demand service with autonomous vehicles that is part of normal public transport operations and based on AI. These are to be integrated into normal public transport operations in Darmstadt and in the district of Offenbach. The service can be booked via an app.

The first autonomous shuttles have been gradually introduced in test operations since June 2024. Technical supervisory staff is monitoring the driving manoeuvres of the vehicles. The technology allows them to safely share the roads with normal traffic. In the future, the shuttles are to be operated on the roads without safety drivers.

Use case: NeMo.bil project – Development of a system of cooperating vehicles for sustainable, individualized public transport

The NeMo.bil research project is developing and prototyping an innovative, swarm-like mobility system with a modular, autonomous shuttle fleet in rural regions. This enables a new form of sustainable and needs-based passenger and goods transport in rural areas.

In 2026, compact, scalable, electrified and very light vehicles (NeMo.Cabs) are to be used as prototypes for autonomous transport on the first and last mile in public space with safety drivers. In the future, NeMo.Cabs are to provide connected on-demand mobility services for individualized public transport in swarms; over longer distances, they are to be towed by larger towing vehicles (NeMo.Pro) in a convoy and charged during the journey. Neue Mobilität Paderborn is a registered association that researches user acceptance and networks stakeholders from local authorities in the Paderborn area.

MINGA

Use case: MINGA project – Automation of vehicles in local public transport

In the MINGA research project, numerous partners are pursuing the goal of evolving local public transport with ride pooling, solo buses und bus platoons under the coordination of the City of Munich.

By the end of 2025, research will be carried out into how driverless transport systems can be intelligently connected with each other and appropriately integrated into the existing local public transport network. Several autonomous vehicles are planned for use in on-demand operation. The vehicles have to prove themselves in challenging traffic situations, for example in rush-hour traffic or in busy urban neighbourhoods.

In addition, buses travelling in close succession, known as platoons, and an autonomous solo bus are being tested in real passenger operations and dovetailed with the on-demand services.

SAFESTREAM

Use case: SAFESTREAM project

The project focuses on the development and trialling of a driverless Level 4 overall system for the operation of electric vehicles with autonomous driving features in defined operating areas without a safety driver.

Trialling takes place in the real-world laboratories in Kelheim and Monheim.

ABSOLUT II

Use case: ABSOLUT II project

The project addresses the central problem of replacing the existing safety drivers in the vehicles with stationary technical supervisors in a control centre. A system solution is being created that integrates vehicle, infrastructure, control centre and booking systems as an overall approach. In a next step, an industrially manufactured shuttle vehicle is to be integrated in order to prepare the overall solution for subsequent scaling and mass production.

2.3.3 Freight transport

Carriage of goods is characterized by a wide range of tasks with different stakeholders and a wide range of transport distances and vehicles used.

Due to the varying complexity of the sub-markets, a progressive rollout across sub-segments of the logistics chain is to be expected. For example, there are already initial use cases for autonomous vehicles to supplement conventional carriage of goods in freight hubs, such as in large industrial zones or port facilities. In a next step, increased use of autonomous vehicles for shuttle services on defined routes is conceivable, such as between two operating centres, for inland traffic to and from seaports or between logistics hubs. They could also become integral to city life in urban areas, too, with some deliveries being made autonomously. In the long term, autonomous vehicles could also cover large distances for long-haul national and international transport with a few stops and energy-efficient driving programmes.

ATLAS-L4

Use case: ATLAS-L4 funding project for autonomous trucks

In the ATLAS-L4 project, MAN Truck & Bus SE and other partners are demonstrating that Level 4 autonomous vehicles can be used on motorways to connect logistics centres and build the basis for innovative transport and logistics concepts.

The aim of the project is to investigate the contribution to greater safety and less congestion on the motorway. The vehicles are to be operated more fuel-efficiently and the shortage of drivers is to be countered by eliminating less attractive driving tasks.

Initial tests are already taking place on motorways, with further projects between logistics and transport hubs to follow from 2025.

Use case: AUTOGVZ project for automated and remotely supported HGV transport in the Bremen freight village

The AUTOGVZ project in the Bremen freight village aims to apply the technological advances of recent years to freight transport. The project objective focuses on existing HGV shuttle services with tractor units and container chassis in the Bremen freight village. Every day, empty containers are transported from the Roland Umschlag combined transport (CT) terminal to two logistics centres in the Bremen freight village by public road. There, the containers are loaded and transported back to the CT terminal.

The two routes have a length of approx. 3 km. Around 10 classic HGV tractor units are currently used for this purpose. In the use case, vehicles are to be replaced by automated and remotely supported HGV tractor units. The knowledge gained from the project is to be used to also roll out the system in other freight transport centres.

2.4 A look at other modes of transport

2.4.1 Rail

On the railways, the International Association of Public Transport (UITP) distinguishes five grades of automation (GoA), from 0 'On-sight operation' to 4 'Fully automatic train operation'. However, operation at GoA Grade 4 does not mean autonomous operation, as the trains are still controlled via an operations or control centre. A distinction is also made between 'open' railway and tram systems on one hand, and 'closed' metro systems on the other.

Driverless trains are currently primarily used in fully automatic (closed) underground railways and rapid transit systems (e.g. in Nuremberg and Hamburg).

Use case: Hamburg digital rapid transit system

As part of Deutsche Bahn AG's Digital Rail Germany programme, a 23-km section of track in the Hanseatic City of Hamburg has been equipped to drive forward automation in train operations. During the highly automated journey, the train operates at GoA 2. The journey runs automatically from start to finish once it has been initiated by the driver. Train provision is implemented at GoA 4. After all passengers have disembarked, the rapid transit railway turns and runs fully automatically in a reversing track. This rapid transit railway has been used in normal operations since September 2022 as part of the regular transport system.

By 2030, the City of Hamburg is planning to fully digitalize the rapid transit system as part of the Hamburg-Takt. This is to increase the frequency of trains by up to 30%. The trains are also to run more punctually and be more energy-efficient. The long-term goal is the complete digitalization of local public transport as part of a range of sustainable urban mobility services.

Other test operations are underway to develop sensors and detection as a requirement for autonomous operation. Given its guided nature, rail transport appears predestined for autonomous technologies. The practical implementation of automated operation is fundamentally easier for rail-bound systems than for road-based vehicles, especially in closed systems.

Use case: Erzgebirge digital test bed

In the Erzgebirge (Ore Mountains) region, on a stretch between Annaberg-Buchholz and Schwarzenberg, there will be a digital route network and a research platform equipped with state-of-the-art 5G communication technology.

State-of-the-art railway and mobility technologies such as autonomous driving, digital control and signalling technology and real-time data transmission are being tested under real-life conditions. This is a joint project involving various partners from the scientific community, industry and the public sector, including Chemnitz University of Technology and Deutsche Bahn.

Use case: AutomatedTrain project – Fully automated driving on the railways

The AutomatedTrain project aims to demonstrate the technical feasibility of fully automated driving for the use case of fully automated provision and stabling of heavy rail at limited speeds.

The aim is to achieve GoA 4. Particular attention will be paid to the modularity and interchangeability of the systems, parts and technical components to be developed and their suitability for integration into different rolling stock.

2.4.2 Waterways

For international inland waterway transport, the Central Commission for Navigation on the Rhine (CCNR) has also defined six automation levels based on the SAE levels. The 6th level is autonomous navigation. The International Maritime Organization (IMO) now distinguishes four automation levels.

IMO is currently developing international regulations for autonomous seagoing vessels in working groups under the heading 'MASS' (Maritime Autonomous Surface Ships). A regulatory framework is also to be developed at national level for the maritime sector (for ships that are not subject to IMO guidance).

Ships and ferries on waterways, in sea ports and at sea are proving in project and test operations that fully automated operation is already possible (with control personnel on board) and can be incorporated in ongoing non-automated operations. Preliminary stages of partial automation are already standard today.

Use case: CAPTN project in Kiel

The long-term objective of the CAPTN project is establishing a clean autonomous public transport network, including autonomous ferry operations.

The initial goal was to develop an autonomous passenger ferry that would connect the east and west shores of the Kiel Fjord – and thus the university campuses. The CAPTN project aims to create a future-proof transport network. New mobility concepts are being developed, different modes of transport are being connected and the use of sustainable energy is being researched.

2.4.3 Aviation

In Advanced Air Mobility (AAM), a new mode of transport is to be established that can also serve difficult to access regions. Urban mobility, as well as transport of goods and passengers in rural areas, will increasingly feature automated and connected Unmanned Aircraft Systems (UAS) and electric Vertical Take-Off and Landing Aircraft (eVTOL) as a result of new aircraft and technological developments. eVTOL and larger transport drones can relieve the strain on transport infrastructure, are quieter than most other modes of transport and largely zero-emission.

AAM is to be rolled out initially with a pilot on board on predefined routes. As soon as safe operation is demonstrably guaranteed, remotely piloted operation of eVTOL can be assessed. Once the technological prerequisites are advanced enough to ensure safe operation, automated operation of eVTOL is the objective. Spillover effects, for example to flight control systems, collision avoidance in the air sports sector, or high-efficiency battery cell systems are likely to lead to innovations in other transport technologies and in traditional aviation and further enhance their aviation safety and efficiency.

Use case: LabFly project – UAS in medical logistics

The aim of the project is to develop the implementation of drone-based transport of medical samples in order to provide the best possible medical care outside of metropolitan regions. The project serves to prepare and train the normal operation of drones for the purpose of transporting medical samples between hospitals and laboratories over long distances in a region with poor infrastructure.

Autonomous flight beyond visual line of sight is to be trialled as part of the project.

2.5 Multimodal transport

Major developments are also evident in multimodal transport. For instance, in the logistics sector, freight container handling at the Port of Hamburg has been highly automated for several years, with the modes of transport dovetailing to form a multimodal logistics chain. In closed systems (e.g. ports, factory sites), real world operations show that hub-to-hub projects can be implemented across multiple modes of transport.

Integrated and holistic logistics strategies (e.g. logistics centres connected to highways), combined with different modes of transport and their networking reduce turnaround times and permit optimal use of resources. The use of autonomous vehicles allows transport operations to be mainstreamed without any staff whatsoever.

Use case: Automated and cross-modal freight handling at the Port of Hamburg (HHLA AG)

In the Port of Hamburg, autonomous transport vehicles and remote-controlled container gantry cranes have been used for cargo handling for years and are being continuously upgraded. The Container Terminal Altenwerder (CTA) uses autonomous vehicles (Automatic Guided Vehicles, AGV) and AI-supported gantry cranes.

The overall system is increasingly incorporated in intermodal logistics transport. The Hamburg TruckPilot project successfully tested autonomous loading and unloading as well as the integration of autonomous HGVs into the automated container handling process between 2018 and 2021. With the MODI project, Hamburg is also testing the first autonomous HGV transport operations from the Federal motorway A 7 to the terminal site (planned duration 2022-2026).

To a minor extent, these developments are already apparent in human-based transport, e.g. in the form of shared mobility services embedded in transport systems (see the text box below).

ALIKE

Use case: Alike project

The Alike research project is the first to establish an overall system for booking autonomous ride-pooling services in local public transport. The initial focus is on the integration of up to 20 autonomous vehicles. The autonomous on-demand service is to be trialled in real-life operation and will allow digital booking via apps that pool all relevant mobility services on one platform. The project results form the basis for the future commercial provision of ride-pooling services.

Hamburg as a Metropolitan Model Region for Mobility

At the end of 2022, the Federal Ministry for Digital and Transport and Hamburg's Ministry of Transport and New Mobility agreed to cooperate more closely on mobility projects in the future and concluded a declaration of intent on Hamburg as a Metropolitan Model Region for Mobility.

The aim is to develop and build a completely new, digitalized and connected urban mobility system. In the Metropolitan Model Region for Mobility, projects with other countries, the European Commission, industry and the scientific community are to be identified and implemented. The results are to be transferrable to other regions through networking and knowledge transfer.

Digression: Federal motorway A 7 – Model route pilot project

As part of the widening of the Federal motorway A 7 to 8 lanes and the construction of the new motorway A 26 in Hamburg, the telecommunications systems are also being renewed. In this context, a passive infrastructure is being built with the A 7 model route. This makes it possible to react flexibly to future infrastructure requirements for autonomous driving without costly civil engineering work.

3 Targets

Road traffic is an important area of application for innovative, autonomous mobility services. In the future, many people will probably be able to experience the technology in local public transport. The resulting enhancement and increase in flexibility will have a positive impact on the transport system.

As an export-driven economy in the heart of Europe, logistics is also of great importance and helps industry and trade exchange goods across all modes of transport. The technology is to be used nationwide to enable Germany to become **world leading in autonomous freight transport**.

We want the German **automotive industry** to be able to build on its successes to date and maintain its **global lead through innovation**. This includes a widespread rollout of autonomous and connected driving across all areas of application. Developing and implementing responsive services will also increase the demand for autonomous vehicles and boost the automotive industry's innovation and technology leadership in Germany. As part of this process, we are monitoring the impact on the environment, climate and society and taking that into account for the frameworks.

Against this background, the Federal Government has set itself the following goals:

1. **By 2025**, with the federal states, we aim to identify opportunities to promote and finance autonomous mobility services in local public transport and discuss the required fundamentals.

2. We will strive to seamlessly transition autonomous driving from trialling to **normal operations by 2026**. In the future, autonomous transport services are to be a common sight in everyday traffic on the roads.

3. We want to help companies in Germany to develop and introduce competitive solutions and mobility services for local public transport **by 2027**, which will in the long term permit economically self-sufficient deployment integrated in the public transport system.

4. **By 2028**, we want to create the world's **largest coherent operational area** for autonomous vehicles, especially with a focus on connecting suitable stretches of the motorway network with cities and metropolitan regions.

5. **By 2030**, we want to integrate autonomous driving as a permanent part of a multimodal and interconnected mobility system. **Public transport** makes a key contribution to the delivery of a high-performance, reliable and climate-friendly passenger transport system. Appropriate integration of autonomous vehicles in mobility systems allows convenient, affordable and accessible transport services. The aim is to **add appreciable value for society and the environment, and increase the citizens' quality of life**. In this way, social acceptance for autonomous and connected driving can be ensured.

6. We aim to make Germany one of the **world's leading innovation and production hubs for autonomous and connected driving**. The market share of autonomous driving is to rise steadily, especially in local public transport and logistics.

7. Germany is home to a **strong automotive industry** and an **innovative environment for science and research**. German products are known for their quality and used worldwide. To ensure that this remains the case, **we are bringing together all stakeholders in the autonomous and connected driving sector** – including the Federal Government, federal states, local authorities, industry, transport operators, mobility providers, users and the scientific community. The focus is on the evolution and application of artificial intelligence and production of autonomous vehicles by the automotive industry.

8. By enhancing **communication** and **cooperating** more closely with our **international partners**, we will support the establishment of globally uniform framework conditions. Through our commitment for harmonized regulations and standards, we are ensuring secure technologies, fair competition, strengthening Germany's status as an industrial nation and contributing to the evolution of technology.

4 Potential and challenges

Autonomous driving is a key technology on the road to sustainable, safe, digital, accessible and affordable mobility. When establishing autonomous and connected driving in Germany, we must tap the potential without losing sight of the challenges.

Autonomous driving **has the potential** to:

- **increase road safety** through the use of reliable on-board technologies, for example with vehicle-to-everything communication (V2X technology). This technology complements the existing sensors and camera systems as a major component of environmental perception. The use of early warning systems and targeted coordination of vehicles in hazardous situations can further increase road safety. This can reduce both the frequency and severity of accidents.

- **strengthen Germany's status as an innovative and business-focused economy** by developing advanced and economical mobility solutions, by producing autonomous vehicle technologies in Germany and through the corresponding market opportunities and business models.

- increase access to and the attractiveness of local public transport compared with private motorized transport by **expanding the range of mobility service**. This includes innovative autonomous and accessible services such as autonomous shuttles and on-demand solutions that improve **social inclusion** in particular also in rural regions.

- **increase traffic efficiency** by networking optimally coordinated transport operations, flows of traffic and modes of transport in normal operations and reducing congestion. This can **reduce mobility-related emissions** via more efficient infrastructure use, better flows of traffic, flexible and shared vehicle use and improved connectivity with other modes of transport.

- **evolve communal forms of mobility like car sharing**, as well as transport services like **ride-pooling**[6] and **ride hailing**[7] trough autonomous vehicle technologies and, in doing so, **strengthen and diversify the range of mobility services**. That is true for both cities large and small, as well as for rural regions.

---

6 Ride-pooling: Digital passenger requests are grouped and combined to form the most efficient route possible. Passengers are collected successively and brought to different destinations, using virtual stops where appropriate (private fare-based journeys within the meaning of regular on-demand services in accordance with section 44 of the Carriage of Passengers Act (PBefG)).

7 Ride hailing ('robotaxis'): Vehicles that transport one or more passengers from A to B along flexible routes in the form of single trips based on a booking. They can be ordered on-demand via digital booking systems (private fare-based journeys within the meaning of regular on-demand services in accordance with sections 46 and 47 of the Carriage of Passengers Act (PBefG)).

- **help tackle the shortage of driving staff** through driverless systems**.**

- **facilitate more efficient freight transport** by extending driving times in autonomous operation and via integrated logistics strategies (e.g. logistics centres connected to highways) by combining various modes of transport and intermodal interconnection to reduce turnaround times as well as for optimal use of resources.

- support companies along the value chain in providing autonomous mobility services by scaling the range of services. In this, cooperation between vehicle manufacturers, component suppliers, software producers, infrastructure providers and public administration is a key prerequisite for the development and implementation of **sustainable business models in local public transport**.

If this potential is to be tapped, technological, application-specific and geographical **challenges** must be identified and addressed, building on existing strengths. Irrespective of the potential described here and the progress already made, autonomous driving has yet to make the transition to normal operations. There are several reasons for this.

- **Autonomous vehicles ready for mass production** have yet to be launched on the market. Currently, there are no mass-produced vehicles available anywhere in the world. Due to the low production volumes and high development and investment costs, it is not yet economically viable for companies to offer production vehicles. German companies have extensive expertise in vehicle manufacturing, which has to date primarily been applied in the context of driver assistance systems and automated driving in private motorized transport. German stakeholders are driving innovation in this area. However, they have yet to do so for autonomous driving systems.

- Furthermore, the **transformation of the automotive industry** is in full progress. The industry in Germany now not only faces a transition to low-emission drivetrains, but also a shift in value creation from hardware-focused to software-focused. Both of these trends tie up investment and development capacities.

- EU rules governing who can use **vehicle data** and how they can do so (regardless of the level of automation) are in development, but have yet to be finalized.

- **Resilient mobile communications infrastructure and satellite communication** are vital for secure communication. Due to the ever-changing technology components, the **mobile communications and road infrastructure** is also evolving all the time. However, the fundamental goal should be to avoid additional infrastructure requirements for autonomous driving, or at least to minimize them.

- Global advances in automated driving show the fundamental interest, both among the population as a whole and throughout the market in general, including industry. As **public acceptance** plays an important role, the evolution of autonomous driving must focus on the users' perception of safety.

- **High safety standards** and system protection are key enablers for the introduction of autonomous driving. The approval process in Germany (and Europe) is subject to strict overall requirements to ensure the maximum possible safety of the vehicles in real world conditions.

- On the one hand, **connectivity and the digital transformation** are foundational for autonomous driving, while on the other, they present new risks in the form of cyber threats. These must be minimized where possible. As a result, the strict automotive engineering requirements must be supplemented with strict **cyber security requirements**, which the vehicles must fulfil to fend off potential manipulation or external attacks and allow them to be investigated.

A look at other modes of transport

- The **legal basis for automated freight and rail transport** needs to be further refined. For example, driverless operation of tramways is not yet provided for in section 53 of the Regulations on the Construction and Operation of Tramways (BOStrab). Driverless operation is also not yet permitted in freight-based rail transport (Construction and Operation of Railways Regulations, EBO 2019).

- With a view to **automated rail transport** (e.g. closed metro systems), **significant investment** is required in some cases for the development **of the infrastructure** leading up to qualification. **The same applies to inland waterway transport.**

- The **technological challenges** of **safely introducing autonomous technologies in shipping** are **enormous due to the complexity of the systems involved.** The entire system needs to be reorganized in order to adequately and safely ensure the connectivity of ship- and shore-based systems. **The international, European and national regulatory framework** is currently being geared towards facilitating the necessary paradigm shifts.

5 Action areas and measures

We will implement the action areas described below with concrete measures to work towards the goals.

5.1 Evolving the regulatory framework

We will continuously evolve the national regulatory framework without favouring specific technologies.

- Autonomous driving technology and the applications and business models that build on it are ever-evolving. We therefore view adapting the regulatory framework as a continuous process. We check on a regular basis whether, and if so, which adaptations are required, taking into account the requirements of environmental protection, climate change mitigation and road safety. For example, up-to-the-minute experience will inform the approval practice for autonomous vehicles, driving the scalability of the technology for use in normal operations.

- Where necessary, we will adapt road traffic law to define and regulate legal aspects of driver behaviour. Among other things, this includes transferring driver duties to technical supervision and operating staff, along with accident safety obligations that militate against the deployment of driverless vehicles. We will also evaluate the professional qualifications required for technical supervision.

- When evaluating the Act on Autonomous Driving, we will consider its practical viability.

- Lessons learned from the application of the Ordinance on the Approval and Operation of Motor Vehicles with Autonomous Driving Functions in Defined Operational Areas (AFGBV) in practice and other findings will be used to continuously assess the effectiveness of the Ordinance and adapt it if necessary.

We will champion the development of the regulatory framework for autonomous driving at EU level.

- Conditions must be put in place throughout the EU to facilitate a ramp-up of the market for autonomous vehicles and their availability, especially in local public transport. That is why we are specifically championing the extension of the regulatory framework from small to large series, and want to put in place the conditions for approval of large series production.

- We are committed to ensuring fair conditions of competition between vehicle manufacturers and service providers as well as transparent and standardized frameworks for the exchange of mobility data. This data exchange must be organized in compliance with the National Security Strategy and regulatory data protection frameworks. Clear regulations are essential for access to as well as use and secure processing of vehicle data.

- In the EU regulatory processes, we will champion guidance that permits continuous evolution of the technology and its practical and demand-driven deployment based on German law as a precedent.

We are driving forward the process to update and harmonize the international regulatory framework at UNECE[8]-level.

- Autonomous driving does not stop at national borders. Manufacturers need internationally harmonized and reliable regulations for efficient scaling. Harmonized regulations are also the foundations for functioning autonomous cross-border traffic. We will continue to champion harmonization of the regulatory framework in international bodies, such as in UNECE working parties WP.1 (Global Forum for Road Traffic Safety), WP.29 (World Forum for Harmonization of Vehicle Regulations) and GE.3 (Expert Group on drafting a new legal instrument on the use of automated vehicles in traffic) and the Working Party on Automated/Autonomous and Connected Vehicles.

We are putting in place practical and flexible approval processes.

- Our goal is to establish the most comparable and unbureaucratic approval and registration processes. This also includes uniform testing standards for technical services (e.g. TÜV and Dekra in Germany) and simplified recognition of comparable areas based on already approved operational areas. We will assess any process uncertainties and minimize them in a targeted and sustained manner.

- To increase planning certainty, we will – in the event of delays – assess whether deadlines for approval periods for operational areas or vehicles can be brought forward.

- As the market for autonomous driving in the logistics sector ramps up, we will assess whether the location-specific operational area approvals are a barrier to scaling.

- We are assessing simplifications of the requirements for transitioning trial approvals to full approvals.

8 United Nations Economic Commission for Europe

No.	Measure	Parties involved	Time horizon
1.1	Evolve national regulatory framework on autonomous driving (Act on Autonomous Driving, AFGBV, StVG)	Federal Ministry for Digital and Transport	Ongoing task
1.2	Consider environmental protection and climate change mitigation	All parties involved	Ongoing task
1.3	Harmonize and evolve the international regulatory framework (EU, UNECE)	BMDV, KBA, BASt	Ongoing task
1.4	Increase cooperation between federal states and Federal Government to develop uniform testing standards and facilitate mutual recognition	Federal Government, federal states, KBA, BASt, standardization organizations	Ongoing task
1.5	Put in place flexible and practical approval processes	Federal Ministry for Digital and Transport	Ongoing task

5.2 Making Germany more attractive for business and innovation

In recent years, German automotive manufacturers and component suppliers have made major leaps forward in innovation, and successfully positioned more and more automated technologies on the market. Strengthening research, innovation and technology development in Germany is vital for its future success in international competition.

- We will identify investment and technology incentives for domestic production of the necessary enabling technologies and the vehicles and implement them with suitable measures. The aim is for national and international manufacturers to produce for the global market in Germany. For competition and innovation, the market needs multiple providers of autonomous driving technologies.
- In order to allow the world's largest coherent operational area for autonomous vehicles to be created, we will promote networking of the stakeholders involved.

In order to continue to enhance Germany's status as a technology hub for autonomous driving, we will strengthen the unique research environment.

- We will ensure that research funding remains focused on applied research and that the resulting findings make a lasting transition to practical application. To this end, we will update the 'Research for Autonomous Driving' Action Plan and align it with specific research priorities.

- Research cooperation between the automotive industry, component suppliers, infrastructure managers, universities, research establishments, associations and authorities is to be enhanced. Sharing laboratories and facilities can minimize costs and risks and generate synergies. This will allow potential gaps in research and test infrastructure to be identified early and rectified. We also want to promote pre-competitive cooperation of stakeholders in the research community.

- Successful approaches from Federal funding projects will be compiled as best practices, including necessary incentives, instruments and success criteria, and made available to other stakeholders. This increases the prospects for implementation in normal operations after the completion of funding and improves networking between the funding recipients.

- The requirements for sensors, infrastructure interfaces and interaction and cooperation with other road users are already being tested in trial environments that are largely independent of one another (including scenario-based trials). We will strive to ensure that both their expansion and their use are coordinated. We also advocate and support an at least Europe-wide uniform standardization and definition of the interfaces.

We will effectively counter the expected shortages of specialist drivers with new qualification requirements in initial and continuing training.

- Introducing autonomous forms of traffic can help to reduce shortages of professional drivers for road vehicles. At the same time, highly specialized professionals will be needed in other professions. We want to contribute to ensuring that new qualification requirements are incorporated in initial and continuing training, such as integrated traffic planning, digital traffic signals or artificial intelligence. Projects related to initial and continuing training should be promoted, as should higher education STEM programmes.

- In addition, new jobs will arise, for examples in technical supervision, maintenance and repair of autonomous vehicles. Further analysis is required for the development of corresponding job profiles.

No.	Measure	Parties involved	Time horizon
2.1	Examine the use of economic and transport policy investment incentives	BMWK, BMDV	Ongoing task
2.2	Support the creation of the world's largest coherent operating area	BMDV, federal states, local authorities, Autobahn GmbH	2028
2.3	Update the 'Research for Autonomous Driving' Action Plan	BMBF, BMWK, BMDV	2025
2.4	Strengthen cooperation in research	Industry, universities, research establishments, associations, public authorities, economic operators	Ongoing task
2.5	Publish best practices from Federal Government funding programmes	BMBF, BMWK, BMDV, funding projects	2030
2.6	Coordinate the deployment and use of test beds	BMDV, BMWK, test bed operators	2030
2.7	Adapt initial and continuing training to new qualification requirements	Federal Government, universities, chambers, trade unions, industry	Ongoing task

5.3 Integrating autonomous mobility services in public transport

We will transition autonomous driving from trial operations to normal operations. We believe that public transport can drive innovation. We want to establish autonomous driving as an integral component in the range of public transport services.

- The Federal Government continues to fund projects to transition promising applications in passenger and freight transport to normal operations. In this, we will focus on projects that target normal operations and/or are close to market maturity, help to reach the Federal Government's strategic goals and can be deployed with minimal interventions in the existing infrastructure.

- Promoting research and development projects (R&D projects) in the mobility sector facilitates the sustainable development of new business models and technologies that make Germany more attractive as a place to do business compared with other countries.

In this way, R&D promotion makes a key contribution to job creation, economic growth and to future-proofing our economy. With regard to financial assistance programmes, we will advocate a coherent focus of support policy in the Federal Government and with the federal states.

- Cooperation between stakeholders from the scientific community, industry and society is imperative and is strengthened in particular by promoting collaborative R&D projects. Sharing knowledge and resources creates synergies and accelerates the development of innovative technologies and systems.

- We are prioritizing the deployment of autonomous driving features within the existing national regulatory framework (Road Traffic Act (StVG) and AFGBV). Applications in local public transport and freight transport with defined operational areas are being considered. In this way, the complexity of the requirements for autonomous systems and the likelihood of unexpected driving situations can be reduced. A gradual extension of the respective operating areas and use cases in line with the technological advances can limit the economic risk and boost the technology's market ramp-up. Standardized operational areas are to be developed as blueprints for the local authorities.

- In autonomous freight transport, arrangements for handover of goods and documents as well as interfaces to recipients are essential for seamless and efficient supply chains. These tasks are currently performed by drivers. In dialogue with the stakeholders involved we will draw up solutions for new forms of cooperation and derive appropriate measures.

- We are assessing options for the deployment of autonomous vehicles in the fleets of the federal authorities.

We will support federal states and local authorities in integrating autonomous services as efficiently and sustainably as possible across all modes of transport.

- In urban areas, autonomous ride-pooling services are to contribute to improving existing public mobility services and to making them more efficient. In less densely populated rural regions, autonomous ride-hailing services can also ensure the important function of fundamental mobility beyond private motorized transport. As a result, we support the integration of corresponding services in local mobility strategies. This is to promote intermodal and multimodal mobility by integrating public transport, cycling and walking. At the same time, specific measures are dependent on local factors.

- We welcome the creation of model regions, the formation of networks and mutual transfers of technology. The implementation of mobility projects is to be accelerated and supported through ongoing exchange and defined processes. The results will be transferred to other regions and cities.

- The local authorities are to be independently capable of creating and operating autonomous mobility services. To this end, we are expanding cooperation between Federal and federal state authorities and promoting optimum transfers of technology. To the extent legally possible, we will systematically make available the lessons learned in project and real world operations as blueprints (e.g. standardized operational areas) for similar scenarios.

- Together with the federal states and local authorities, we will work on solutions to facilitate procurement and deployment of autonomous vehicles. Guidance on integration of autonomous vehicles in transport strategies and contract specifications could help to apply uniform procurement criteria. This also includes changes to the local transport plans and establishing procurement networks of multiple local authorities.

- We will also start a dialogue of all stakeholders so that issues with implementation are driven forward in a holistic manner. To this end, we are initiating an Implementation Alliance to scale up autonomous on-demand shuttles incorporating all relevant stakeholders from federal states, local authorities, operators, companies, associations and partners in the scientific community. Together, we will identify concrete barriers and needs for better framework conditions and support, and implement them taking the responsibilities and possibilities of the respective stakeholders into account.

- Flexible use of autonomous on-demand services and their integration in existing local public transport systems requires interoperable control systems. We will therefore advocate joint development of open system architectures and joint interfaces. This allows creation of a broad, modular range of services that will permit local authorities and operators to combine solution components by various manufacturers. That promotes competition, which in turn contributes to cost reduction and scaling. In addition, this can help avoid lock-in effects and dependency on individual providers.

- For the implementation of autonomous vehicle systems in local public transport, we are working with the federal states and local authorities to assess the extent to which existing funding systems can be used.

No.	Measure	Parties involved	Time horizon
3.1	Promote use in passenger and freight transport	BMDV, BMWK	2030
3.2	Develop solutions and measures for new forms of cooperation in autonomous freight transport	BMDV, BMWK, industry associations, scientific community	2030
3.3	Examine the possible use of autonomous vehicles by federal authorities	Federal Government	Ongoing task
3.4	Expand networking of Federal Government and federal states to integrate autonomous services into regional transport systems and develop instruments to strengthen intermodal and multimodal mobility	Federal Government, federal states, scientific community, associations	2030
3.5	Initiate an alliance for implementation	Federal Government, federal states, local authorities, operators, businesses, associations and scientific partners	2025
3.6	Examine funding schemes to deliver autonomous vehicle systems in local public transport	BMDV, federal states, local authorities	2025

5.4 Evolving infrastructure and technology

Autonomous and connected driving can make a major contribution to increasing road safety, for example if vehicles receive real-time information on slippery road surfaces or the end of a tailback. We will evolve and harmonize standards for interaction between vehicles and with infrastructure and other road users. This also determines the system security.

- At both European and international levels, we will champion the uniform definition, standardization and harmonization of framework conditions, interfaces and C-ITS services for communication between vehicles (especially also with emergency vehicles) and between vehicles and infrastructure.

- For human-machine communication between autonomous vehicles and vehicle users as well as with other road users, we believe that national and international standardization is necessary. Together with industry, the scientific and research communities and other authorities, we will support and advance their development.

- Due to the absence of on-board staff in the vehicles, user-friendly communication and safety concepts are required. We are supporting their development.

- We also want to ensure the development of a concept to provide a nationwide PKI (Public Key Infrastructure), taking into account valid European and national standards and recommendations to support the deployment of C-ITS services on German roads.

Where necessary, we will actively shape the infrastructure for autonomous and connected driving. We will provide quality-controlled real-time information on the usability of the infrastructure.

- Requirements for road traffic and data infrastructure are to be collected and coordinated on an ongoing basis, and transitioned to suitable measures as required. To this end, we want to bring together and coordinate the Federal Government, the federal states, industry, the scientific community and project operators. The aim is to expand the scope of application of autonomous driving systems and create potential standards for operation, maintenance, construction and alteration of the road infrastructure.

- We will gradually improve the provision of information on currently valid traffic rules and regulations and the usability of the road infrastructure or on the traffic volume via the Mobilithek. In this way, autonomous vehicles (too) can predictively check the probable usability of a road based on the environmental and traffic conditions and take this information into account when making decisions.

We will also continue to drive upgrades to the mobile communications infrastructure in order to create the best possible framework conditions for autonomous driving in Germany.

- High-capacity and stable mobile communications networks support autonomous and connected driving systems. The Federal Government plans to eliminate the remaining white spots, especially along the transport infrastructure, by means of a dynamic, market-driven roll-out. For this purpose, accompanying measures were defined in the Gigabit Strategy and have been and are being implemented successively.

We will take into account the Galileo satellite program and in particular the Galileo services for the autonomous driving use case and examine potential requirements, for example with regard to establishing redundancies in the communications infrastructure, in order to boost the resilience of the systems.

- As part of the Galileo satellite programme, services that deliver high-precision navigation and precise positioning and time information are being developed. Accordingly, the aim of evolving the Galileo services in line with demand is to study areas of application in the autonomous driving sector while also facilitating better protection of the communication systems.

We are supporting scenario-based testing and will create a framework for action to deliver the results to the stakeholders involved in a systematic form.

- Virtual test methods are to be supported by establishing a framework for action with general requirements, standards and research findings in the form of specific metrics, reference models and requirements for test methods.

- The fundamental test scenarios presented in the AFGBV are to be supplemented with application-specific scenarios featuring relevant and particularly critical driving situations. To this end, we will support the establishment of a scenario database. A multi-user body at the Federal Motor Transport Authority (KBA) is to be entrusted to manage and coordinate this.

We will advocate making homologation after software updates to safety-critical driving features more efficient.

- The administrative procedures accompanying the software updates are to be accelerated taking European law into account and utilizing national scope to regulate vehicle changes to approved vehicles. Besides safety-critical updates, this would also allow innovations to be implemented faster. Under supervision of the Federal Motor Transport Authority, a process is to be established to allow companies and relevant neutral assessors to independently provide evidence in the context of corresponding regulations at EU and UNECE levels.

We will continue and enhance the use of artificial intelligence and digital twins in digital areas of application and in demonstration projects.

- Autonomous vehicles could not be made without artificial intelligence. Artificial intelligence plays a key role in both the development of data processing systems for environment perception and also in the development of systems to support decision making. The goal-oriented use of AI must be evolved continuously to improve the safety and efficiency of road traffic and minimize the risk of unforeseen events.

- Generative AI methods and appropriate foundation models will permit scalable and secure solutions to enable the extension of operational areas. This will be key, especially when it comes to training autonomous driving features, as well as for the imitation of human-like driving strategies. The training goal of end-to-end learning is to minimize the deviation between how a human driver would drive and the driving strategies generated by the model. The Federal Government is therefore continuing to promote research and development in cooperation projects involving industry and cutting-edge research.

- Digital twins can support autonomous driving. They deliver a precise representation of the current traffic situation or support the operation of existing infrastructure. We therefore advocate the trialling of digital twin applications for autonomous driving.

We will assess the use of developments from various sectors for autonomous vehicle technology applications to tap additional potential in the global competition.

- Environment perception and interpretation, data security and robust navigation and communication between vehicles and with their surroundings and road users play an important role, for example in security and defence research. We will evaluate the use of relevant developments from the security and defence sector for autonomous driving applications. This also applies to using civil applications of autonomous driving technologies for national defence.

No.	Measure	Parties involved	Time horizon
4.1	Develop harmonized standards for interaction between vehicles, infrastructure and users	BMDV, BMWK, BASt, KBA, industry, research community, scientific community, standardization organizations	2027
4.2	Draw up a strategy for a nationwide C-ITS PKI	BMDV, Autobahn GmbH, BASt, federal states	2025
4.3	Exchange requirements to be met by infrastructure in a structured form	BMDV, federal states, industry, scientific community, project operators	2026
4.4	Improve the provision of information related to roads and traffic via the Mobilithek	BMDV, Autobahn GmbH, BASt, federal states and local authorities, transport operators, data users	Ongoing task
4.5	Investigate areas of application for Galileo services	BMDV, scientific community, industry	2026
4.6	Support virtual test methods by establishing a framework for action	KBA	End of 2026
4.7	Set up a database of scenarios with relevant and particularly critical driving situations	KBA	End of 2026

No.	Measure	Parties involved	Time horizon
4.8	Establish a procedure for legally secure, independent verification after software updates	KBA	Mid-2026
4.9	Evolve the use of AI and application of digital twins	BMDV, industry, scientific community, funding projects	2030
4.10	Investigate the use of developments of various sectors	BMDV, BMWK, BMBF, scientific community, industry	2030

5.5 Promoting adoption

We want to raise awareness among both the population and companies of the potential applications and the benefits of autonomous and connected driving.

- Together with the federal states, industry, associations, the scientific community and research projects, we will continue to actively shape the societal dialogue. The aim is to present the opportunities and potential of autonomous and connected driving for society and to show that we are responsibly managing inherent risks.

- In addition to strengthening Germany as a competitive business location and hub of innovation in the context of research and technology development, we also want to increase adoption of the technology and further allay any reservations about it among companies through dialogue and demonstration of its potential.

- We link research projects with supporting surveys, publications and studies to identify issues with regard to societal adoption of the technology. To enhance societal dialogue, the introduction of autonomous driving will be accompanied by high-profile measures and participatory formats.

- In order to create transparency regarding the opportunities and limits of the technology and to strengthen the adoption of autonomous driving – even in the event of a risk or accident – data on critical traffic situations or accidents should be collected and analysed by an investigation body that is to be set up.

- We will enhance the discourse on the necessary processing of image and audio data from the vehicle environment and the right to informational self-determination of road users. The aim is to increase road safety through automated driving features based on the processing of audio-visual environment data in compliance with data protection requirements.

No.	Measure	Parties involved	Time horizon
5.1	Enhance dialogue	BMDV, BMWK, federal states, scientific community, industry, associations, research projects	Ongoing task
5.2	Collect and analyse data on critical traffic situations or accidents	BASt, KBA	Ongoing task
5.3	Enhance the discourse on the use of audio-visual data	BMDV, associations, industry, BfDI	2027

5.6 Enhancing cyber security and data protection

Together with our international partners, we will continue to work on solutions to boost the exchange of information as well as cooperation on risks to cyber security and data protection risks in autonomous and connected driving.

- The increasing connectivity among vehicles and between vehicles and infrastructure as well as the increasing level of vehicle automation raises the risk of cyber attacks. Due to the elevated threat level worldwide, a joint approach with European and international partners is essential to establish standards for cyber security and privacy-by-design (protection of privacy through technological design).

At national level, we will work towards expanding cooperation on cyber security and data protection between vehicle manufacturers, suppliers, research institutions as well as safety and regulatory authorities.

- The development and implementation of cyber-secure solutions, which also protect the privacy of vehicle users, is essential for the acceptance and use of autonomous vehicles. The vehicles use sensors and cameras to record environment and other sensitive data, interact with the infrastructure and exchange data with each other. Securing the systems and technical interfaces against unauthorized access by third parties and preventing misuse of this technology is and remains an ongoing task.

- The exchange of information on cyber security incidents and vulnerability reports between all parties involved, particularly in the area of autonomous and connected driving, is to be enhanced and expanded.

- Artificial intelligence makes an innovative contribution to improving cyber security. At the same time, however, AI can also be used as a target and tool for attacking automated and connected systems. Not only is it necessary to continuously evolve the technology itself, the technology also needs to be verifiable. Furthermore, both targeted attacks as well as malfunctions need to be prevented by technical means. The use of secure AI that can be operated in compliance with data protection regulations is therefore a joint task for the industry, scientific community and authorities.

We will continuously review existing cyber security and privacy-by-design regulations, standards and recommendations and, where necessary, advocate or adopt appropriate amendments.

- Regulations and standards ensure that autonomous vehicles comply with strict safety requirements and can be operated in compliance with data protection regulations. There are already international guidelines and industry-wide standards that regulate the safety and reliability of autonomous vehicles. The aim here is to continuously adapt these to changing framework conditions.

No.	Measure	Parties involved	
6.1	Enhance cooperation in cyber security and data protection	Industry, scientific community, research establishments, authorities, international partners	2027
6.2	Review and adapt regulations, standards and recommendations	Ministries, authorities, standardization organizations	Ongoing task

5.7 Working towards international harmonization

We strive to cooperate closely with our partners at international level. This takes place in the relevant committees and institutions, but also in the form of bilateral, international cooperation.

- We see the opening of additional markets and improved access to foreign markets as an opportunity. Our aim is the mutual recognition of regulations, the harmonization of standards and an open exchange of experience on the basis of fair competition.
- Germany also cooperates bilaterally and multilaterally with various countries beyond Europe's borders. We will strengthen existing bilateral cooperations with like-minded partners and establish new cooperations with major like-minded trade partners.

We will remain active in international bodies for the establishment of uniform legal and technical requirements for autonomous driving.

- To remain competitive, German car manufacturers and their suppliers need globally applicable regulations. We will leverage multilateral relationships, particularly within the G7, G20, OECD and UNECE, to define and evolve both legal and technical framework conditions for autonomous driving.

No.	Measure	Parties involved	Time horizon
7.1	Intensify collaboration at international level (e.g. through cooperations)	BMDV, BMWK, AA	2030
7.2	Evolve uniform international framework conditions	BMDV, BMWK, BMBF	Ongoing task

5.8 Supporting cooperation among all stakeholders

Autonomous driving can only evolve if all stakeholders work together constructively and have a common understanding of the development status, challenges and necessary measures. Continuous evaluation of the legal and organizational framework conditions is therefore essential if this technology is to establish itself and succeed on the market.

The aim is to swiftly implement the Strategy in order to compete globally as a technology hub for the automotive industry. The Federal Ministry for Digital and Transport has the lead responsibility for the implementation, monitoring and strategic management. The Ministry will coordinate closely with the other government departments. The lead responsibility of other departments for individual areas involved remains unaffected by the implementation of the Strategy.

The Round Table on Autonomous Driving has been supporting the evolution of autonomous driving with its experts from industry, the scientific community and society for many years. It has taken a stand on technical, legal and societal challenges, thereby providing important and necessary impetus. The same is true of the Expert Group on Transformation in the Automotive Industry, an independent advisory body to the Federal Ministry for Economic Affairs and Climate Action, which has developed target- and addressee-oriented recommendations for government, industry and societal action to ramp up the market in Germany. We will continue this work with the relevant departments and augment it through structured specialist dialogues.

We will continuously monitor and evaluate the action areas and measures outlined in this Strategy in established bodies such as the Round Table on Autonomous Driving, in relevant bodies of the Federal Government and the federal states, as well as at State Secretary level.

Summary of the individual measures

No.	Measure	Parties involved	Time horizon
Evolve the regulatory framework			
1.1	Evolve national regulatory framework on autonomous driving (Act on Autonomous Driving, AFGBV, StVG)	BMDV	Ongoing task
1.2	Consider environmental protection and climate change mitigation	All parties involved	Ongoing task
1.3	Harmonize and evolve the international regulatory framework (EU, UNECE)	BMDV, KBA, BASt	Ongoing task
1.4	Increase cooperation between federal states and Federal Government to develop uniform testing standards and facilitate mutual recognition	Federal Government, federal states, KBA, BASt, standardization organizations	Ongoing task
1.5	Put in place flexible and practical approval processes	BMDV	Ongoing task
Make Germany more attractive for business and innovation			
2.1	Examine the use of economic and transport policy investment incentives	BMWK, BMDV	Ongoing task
2.2	Support the creation of the world's largest coherent operating area	BMDV, federal states, local authorities, Autobahn GmbH	2028
2.3	Update the 'Research for Autonomous Driving' Action Plan	BMBF, BMWK, BMDV	2025
2.4	Strengthen cooperation in research	Industry, universities, research establishments, associations, public authorities, economic operators	Ongoing task
2.5	Publish best practices from Federal Government funding programmes	BMBF, BMWK, BMDV, funding projects	2030

No.	Measure	Parties involved	Time horizon
2.6	Coordinate the deployment and use of test beds	BMDV, BMWK, test bed operators	2030
2.7	Adapt initial and continuing training to new qualification requirements	Federal Government, universities, chambers, trade unions, industry	Ongoing task
Integrating autonomous mobility services in public transport			
3.1	Promote use in passenger and freight transport	BMDV, BMWK	2030
3.2	Develop solutions and measures for new forms of cooperation in autonomous freight transport	BMDV, BMWK, industry associations, scientific community	2030
3.3	Examine the possible use of autonomous vehicles by federal authorities	Federal Government	Ongoing task
3.4	Expand networking of Federal Government and federal states to integrate autonomous services into regional transport systems and develop instruments to strengthen intermodal and multimodal mobility	Federal Government, federal states, scientific community, associations	2030
3.5	Initiate an alliance for implementation	Federal Government, federal states, local authorities, operators, businesses, associations and scientific partners	2025
3.6	Examine funding schemes to deliver autonomous vehicle systems in local public transport	BMDV, federal states, local authorities	2025
Evolving infrastructure and technology			
4.1	Evolve harmonized standards for interaction between vehicles, infrastructure and users	BMDV, BMWK, BASt, KBA, industry, research community, scientific community, standardization organizations	2027
4.2	Draw up a strategy for a nationwide C-ITS PKI	BMDV, Autobahn GmbH, BASt, federal states	2025

No.	Measure	Parties involved	Time horizon
4.3	Exchange requirements to be met by infrastructure in a structured form	BMDV, federal states, industry, scientific community, project operators	2026
4.4	Improve the provision of information related to roads and traffic via the Mobilithek	BMDV, Autobahn GmbH, BASt, federal states and local authorities, transport operators, data users	Ongoing task
4.5	Investigate areas of application for Galileo services	BMDV, scientific community, industry	2026
4.6	Support virtual test methods by establishing a framework for action	KBA	End of 2026
4.7	Setup a database of scenarios with relevant and particularly critical driving situations	KBA	End of 2026
4.8	Establish a procedure for legally secure, independent verification after software updates	KBA	Mid-2026
4.9	Evolve the use of AI and application of digital twins	BMDV, industry, scientific community, funding projects	2030
4.10	Investigate the use of developments of various sectors	BMDV, BMWK, BMBF, scientific community, industry	2030
Promoting adoption			
5.1	Enhance dialogue	BMDV, BMWK, federal states, scientific community, industry, associations, research projects	Ongoing task
5.2	Collect and analyse data on critical traffic situations or accidents	BASt, KBA	Ongoing task
5.3	Enhance the discourse on the use of audio-visual data	BMDV, associations, industry, BfDI	2027

No.	Measure	Parties involved	Time horizon
Enhance cyber security and data protection			
6.1	Enhance cooperation in cyber security and data protection	Industry, scientific community, research establishments, authorities, international partners	2027
6.2	Review and adapt regulations, standards and recommendations	Ministries, authorities, standardization organizations	Ongoing task
Commitment to international harmonization			
7.1	Intensify collaboration at international level (e.g. through cooperations)	BMDV, BMWK, AA	2030
7.2	Evolve uniform international framework conditions	BMDV, BMWK, BMBF	Ongoing task

Publication details

Published by
Federal Ministry for Digital and Transport
Invalidenstr. 44, 10115 Berlin

As at
December 2024

Layout | Print
Federal Ministry for Digital and Transport | Ministry Printers

Picture credits
Front/back cover: © Sodapeaw – stock.adobe.com
Page 1: BMDV
Page 6: © metamorworks – stock.adobe.com
Page 11: © Stacy – stock.adobe.com
Page 13: © Chronicles Artsy – stock.adobe.com
Page 16: © M – stock.adobe.com
Page 23: © florynstudio3 – stock.adobe.com

www.bmdv.bund.de

linkedin.com/company/bmdv-bund
facebook.com/bmdv
instagram.com/bmdv
threads.net/@bmdv
x.com/bmdv
bsky.app/profile/bmdv.bsky.social
social.bund.de/@bmdv
youtube.com/bmdv